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Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Arranging data in a specific order
- b. Finding a specific element in a list
- c. Removing duplicates from a list
- d. Merging two datasets

The correct answer is: Arranging data in a specific order

Question **2**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It slows down the process
- b. It makes the data unsorted
- c. It makes it easier to select items based on their relationship to the rest of the items
- d. It complicates the selection of items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **3**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

- a. A sorting algorithm is stable if it preserves the order of duplicate keys
- b. A sorting algorithm is stable if it preserves the order of all keys
- c. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
- d. A sorting algorithm is stable if it preserves the order of non-duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **4**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Merging
- b. Sorting
- c. Complexity
- d. Searching

The correct answer is:
Complexity

Question **5**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Combining sorted sublists
- b. Comparing each element with the others
- c. Sorting the entire list sequentially
- d. Dividing the input into smaller subproblems

The correct answer is: Dividing the input into smaller subproblems

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Merge Sort
- b. Bubble Sort
- c. Binary Search
- d. Linear Search

The correct answer is: Bubble Sort

Question **7**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It requires external libraries
- b. It is less efficient than custom sorting algorithms
- c. It only works with integer arrays
- d. It sorts data out of the box efficiently

The correct answer is: It sorts data out of the box efficiently

Question **8**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It is based on the divide-and-conquer approach
- b. It sorts data using a single pass
- c. It is less efficient than bubble sort
- d. It does not require recursion

The correct answer is: It is based on the divide-and-conquer approach

Question **9**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Linear Search
b. Binary Search
c. Merge Sort
d. Bubble Sort

The correct answer is: Merge Sort

Question **10**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Insertion
b. Distribution
c. Extraction
d. Selection

The correct answer is:
Insertion

Question **11**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It has no effect on the search operation
b. It allows for faster searching
c. It makes the search operation slower
d. It increases the number of comparisons needed

The correct answer is: It allows for faster searching

Question **12**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Selection
b. Searching
c. All of the above
d. Duplicates

The correct answer is: All of the above

Question **13**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Merge sort
b. Selection sort
c. Quick sort
d. Heap sort

The correct answer is:
Merge sort

Question **14**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. The parts are merged without sorting
b. Each part is sorted independently
c. The parts are ignored
d. Each part is searched for a specific element

The correct answer is: Each part is sorted independently

Question **15**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Insertion Sort
b. Merge Sort
c. Bubble Sort
d. Quick Sort

The correct answer is: Merge Sort